EE1007
Introduction to Computer Science Laboratory

Lab 9
Vector and String

Andy, Yu-Guang Chen
Assistant Professor, Department of EE
National Central University
andyygchen@ee.ncu.edu.tw

Lab 9.1
Uppercase String

- In this problem, you need to write a program which can firstly capitalize the first letter of each vocabulary in the given sentences, and combine the two sentences to form a single sentence.
- In the beginning, the program will take two sentences which only contains lowercase letters, spaces, and special characters (i.e., !, @, #, …) as inputs. Each input sentence will contains no more than 50 characters. Here we assume all vocabularies will start from a lowercase letter.
- The program will capitalize the first letter of each vocabulary in the sentences, and then combine the two input sentences to form a single sentence. Finally, the program will display the single sentence.

Lab 9.1
Uppercase String

- The screen dialog should appear as follows:

```
Input Line1: an apple a day
Input Line2: keeps the doctor away
Output Line: An Apple A Day Keeps The Doctor Away

Process returned 0 (0x0)   execution time : 28.725 s
Press any key to continue.
```


Lab 9.1
Uppercase String

- You CANNOT use any function which can convert lowercase to uppercase in either C++ standard library or copy from internet. Otherwise, you will get 0 point.

- Hint: You can use the ASCII Table to help you complete the program.

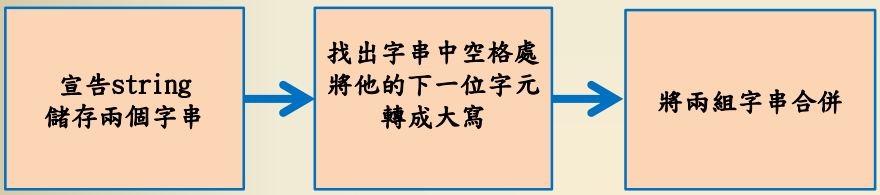
Hint: ASCII Table

ASCII Table

Dec	Hex	Oct	Char	Dec	Hex	Oct	Char	Dec	Hex	Oct	Char	Dec	Hex	Oct	Char
0	0	0		32	20	40	[space]	64	40	100	@	96	60	140	`
1	1	1		33	21	41	!	65	41	101	A	97	61	141	a
2	2	2		34	22	42	"	66	42	102	B	98	62	142	b
3	3	3		35	23	43	#	67	43	103	C	99	63	143	c
4	4	4		36	24	44	$	68	44	104	D	100	64	144	d
5	5	5		37	25	45	%	69	45	105	E	101	65	145	e
6	6	6		38	26	46	&	70	46	106	F	102	66	146	f
7	7	7		39	27	47	'	71	47	107	G	103	67	147	g
8	8	10		40	28	50	(	72	48	110	H	104	68	150	h
9	9	11		41	29	51	)	73	49	111	I	105	69	151	i
10	A	12		42	2A	52	*	74	4A	112	J	106	6A	152	j
11	B	13		43	2B	53	+	75	4B	113	K	107	6B	153	k
12	C	14		44	2C	54	,	76	4C	114	L	108	6C	154	l
13	D	15		45	2D	55	-	77	4D	115	M	109	6D	155	m
14	E	16		46	2E	56	.	78	4E	116	N	110	6E	156	n
15	F	17		47	2F	57	/	79	4F	117	O	111	6F	157	o
16	10	20		48	30	60	0	80	50	120	P	112	70	160	p
17	11	21		49	31	61	1	81	51	121	Q	113	71	161	q
18	12	22		50	32	62	2	82	52	122	R	114	72	162	r
19	13	23		51	33	63	3	83	53	123	S	115	73	163	s
20	14	24		52	34	64	4	84	54	124	T	116	74	164	t
21	15	25		53	35	65	5	85	55	125	U	117	75	165	u
22	16	26		54	36	66	6	86	56	126	V	118	76	166	v
23	17	27		55	37	67	7	87	57	127	W	119	77	167	w
24	18	30		56	38	70	8	88	58	130	X	120	78	170	x
25	19	31		57	39	71	9	89	59	131	Y	121	79	171	y
26	1A	32		58	3A	72	:	90	5A	132	Z	122	7A	172	z
27	1B	33		59	3B	73	;	91	5B	133	[	123	7B	173	{
28	1C	34		60	3C	74	<	92	5C	134	\	124	7C	174	|
29	1D	35		61	3D	75	=	93	5D	135	]	125	7D	175	}
30	1E	36		62	3E	76	>	94	5E	136	^	126	7E	176	~
31	1F	37		63	3F	77	?	95	5F	137	_	127	7F	177	

Lab 9.1
Uppercase String

◆思考方向：



Lab 9.1
Uppercase String

◆Step. 1

```
string line1;
string line2;
int i = 0, j = 0;

cout << "Input Line1: ";
getline(cin, line1);
cout << "Input Line2: ";
getline(cin, line2);
```


Lab 9.1
Uppercase String

◆Step. 2

```
line1[0] -= 32;
while( line1[i] != '\0' )
{
    if( line1[i] == ' ' )
        line1[i+1] -= 32;
        i++;
}

line2[0] -= 32;
while( line2[j] != '\0' )
{
    if( line2[j] == ' ' )
        line2[j+1] -= 32;
        j++;
}
```

Lab 9.1
Uppercase String

◆Step. 3

```
line1 += ' ';
line1 += line2;

cout << "Output Line: " << line1 << endl;
```



Lab 9.2
Vector Sorting

◆In this problem, you need to write a program which can sort the given numbers and find their median.

◆In the beginning, the program will ask you to enter a number which represents the size of the vector, and follows by corresponding integers as given numbers.

◆Then the program will sort given numbers in increasing order.

◆The range of each input integer is $[-2^{31}-1, 2^{31}-1]$.

◆Finally, the program will find the median of these numbers and output the median.



Lab 9.2
Vector sorting

- If there are even numbers in the sequence, the median is the average of the middle two numbers, and please round down to integer.

- You need to use vector to store the values, and any existing function (Ex. sort) is not allowed to use. Otherwise, you will get 0 point.

Lab 9.2
Vector sorting

- The screen dialog should appear as follows:

```
Enter a number:
6
1 3 4 2 6 7
After sorting:
1 2 3 4 6 7
Median value of this series is: 3

Process returned 0 (0x0)   execution time : 9.307 s
Press any key to continue.
```

Lab 9.2
Vector sorting

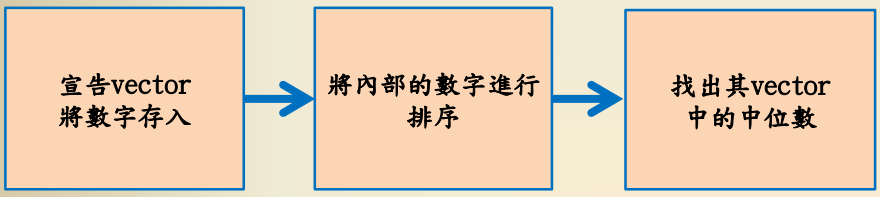
◆ The screen dialog should appear as follows:

```
Enter a number:
5
5 4 4 2 7
After sorting:
2 4 4 5 7
Median value of this series is: 4

Process returned 0 (0x0)   execution time : 15.088 s
Press any key to continue.
```


Lab 9.2
Vector sorting

◆思考方向：



Lab 9.2
Vector sorting

◆Step. 1

```
cout << "Enter a number: " << endl;
cin >> num;

for(int i=0; i<num; i++) //fill the vector
{
    cin >> x;
    line.push_back(x);
}
```


Lab 9.2
Vector sorting

◆Step. 2（此處是使用select sorting）

```
for(i=0; i<num-1; i++) //sorting
{
    for(j=i+1, k=i; j<num; j++)
    {
        if(line[j] < line[k])
            k = j;
    }
    temp = line[k];
    line[k] = line[i];
    line[i] = temp;
}
```

Lab 9.2
Vector sorting

◆Step. 3

```
if(num%2 == 0)
    median = (line[(num/2)-1] + line[num/2])/2;
else
    median = line[(num/2)];

cout << "\nMedian value of this series is: " << median << endl;
```


Lab 9.3
NCU Theater Reservation System(I)

◆In order to provide students more choices to relieve stress, the Principal decided to set up NCU theater for NCU students.

◆NCU theater opened in this December, it attracted a large number of students to watch the movies.

◆For reducing the crowds in line, you are asked to develop a convenient online reservation system.

Lab 9.3
NCU Theater Reservation System(I)

- As a start-up reservation service, the NCU theater only provide 15 seats for students to reserve.
- The status of each seat contains: **available** (default status) and **reserved**.
- For the beginning system, you are asked to develop for functions: 1)Make a new reservation, 2)Cancel an existing reservation, 3) Show the all available seats, 4)Show the status of all current seats, and 5) Exit the system

Lab 9.3
NCU Theater Reservation System(I)

- The main menu of your system is shown as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to exit the system
Enter your selection Here:
```

Lab 9.3

NCU Theater Reservation System(I)

- 1) Make a new reservation
  - First, the system will ask the customer if he/she needs a specified seat or not.
  - If the customer chooses yes, then the system will ask he/she to choose a seat which is available. If the chosen seat is not available, show the message **"The seat has been reserved, please enter your selection again: ".**
  - If the customer chooses no, then the system will find an available seat **which the seat number is minimum** for the customer.
  - A seat can only be assigned to one customer at a time.

Lab 9.3

NCU Theater Reservation System(I)

- 2)Cancel an existing reservation
  - The system will ask you to enter the reservation seat which you want to cancel.
  - If the seat status is reserved, then the status of the seat is change to be available.
  - Otherwise, showing the error message "**The seat is not reserved**" and your program will **return to the main menu**.

Lab 9.3
NCU Theater Reservation System(I)

- 3) Show the all available seats
  - The system will check the seat which is available, and display all the available seats for the customer.
- 4)Show the status of all current seats
  - The system will show the status of all seats with its number.



Lab 9.3
NCU Theater Reservation System(I)

- 5) Exit the system
  - The system will continuous operate until this option is selected.
  - The system will show "**Thank you for using our system Have a nice day~~**" and end its work.



Lab 9.3
NCU Theater Reservation System(I)

- Requirement:
  - If the user enters a wrong input, show the message **"Invalid input, please enter your selection again:"**
- Hint:
  - You can use an array or vector to store the status of the seats.

Lab 9.3
NCU Theater Reservation System(I)

- The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to exit the system
Enter your selection Here: 1

***Make a new reservation***
Do you need a specified seat? (y/n) a
Invalid input, please enter your selection again: y
Enter the seat number: 1
The seat number 1 is reserved for you...

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to exit the system
Enter your selection Here: 1

***Make a new reservation***
Do you need a specified seat? (y/n) y
Enter the seat number: 1
The seat has been reserved, please enter your selection again: 2
The seat number 2 is reserved for you...
```

Lab 9.3
NCU Theater Reservation System(I)

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to exit the system
Enter your selection Here: 1

***Make a new reservation***
Do you need a specified seat? (y/n) n
The seat number 3 is reserved for you...

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to exit the system
Enter your selection Here: 2

***Cancel an existing reservation***
Please enter the seat number which you want to cancel: 4
The seat is not reserved

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to exit the system
Enter your selection Here: 2

***Cancel an existing reservation***
Please enter the seat number which you want to cancel: 2
The seat number 2 is available now...
```


Lab 9.3
NCU Theater Reservation System(I)

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to exit the system
Enter your selection Here: 3

***Show the all available seats***
seat 2
seat 4
seat 5
seat 6
seat 7
seat 8
seat 9
seat 10
seat 11
seat 12
seat 13
seat 14
seat 15
```

Lab 9.3
NCU Theater Reservation System(I)

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to exit the system
Enter your selection Here: 4

***Show the status of all seats***
Seat number       Status
Seat 1            reserved
Seat 2            available
Seat 3            reserved
Seat 4            available
Seat 5            available
Seat 6            available
Seat 7            available
Seat 8            available
Seat 9            available
Seat 10           available
Seat 11           available
Seat 12           available
Seat 13           available
Seat 14           available
Seat 15           available
                  setw(13)
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to exit the system
Enter your selection Here: 5

Thank you for using our system
Have a nice day~~
```


Lab 9.3
NCU Theater Reservation System(I)

◆思考方向：



Lab 9.3
NCU Theater Reservation System(I)

◆Step. 1

```
while(option!=5)
{
    cin >> option;
    while (option < 1 || option > 5)
    {
        cout << "Invalid input, please enter your selection again: ";
        cin >> option;
    }
    cout << endl;
    if(option==1)
    {
        reservation();
        option = printmenu();
    }
    else if(option==2)
    {
        cancel();
        option = printmenu();
    }
    else if(option==3)
    {
        show_available();
        option = printmenu();
    }
    else if(option==4)
    {
        show_status();
        option = printmenu();
    }

}
cout << "Thank you for using our system\n" << "Have a nice day~~"<< endl;
```


Lab 9.3
NCU Theater Reservation System(I)

◆Step. 2（指令1）

```
while (choice != 's' && choice != 'r')
{
    cout << "Invalid input, please enter your selection again: ";
    cin >> choice;
}
if(choice == 's')
{
    cout << "Enter the seat number: ";
    cin >> seatnum;
    while (seatnum < 1 || seatnum > 15)
    {
        cout << "Invalid input, please enter your selection again: ";
        cin >> seatnum;
    }
    while(seats_status[seatnum-1] == 1)
    {
        cout << "The seat has been reserved, please enter your selection again: ";
        cin >> seatnum;
    }
    seats_status[seatnum-1] = 1;
    cout << "The seat number " << seatnum << " is reserved for you... " << endl;
}
else if(choice == 'r')
{
    for(int i=0; i<15; i++)
    {
        if(seats_status[i] == 0)
        {
            cout << "The seat number " << i+1 << " is reserved for you... " << endl;
            seats_status[i] = 1;
            cout << endl;
            return;
        }
    }
}
```

Lab 9.3
NCU Theater Reservation System(I)

◆Step. 2（指令2）

```
cout << "***Cancel an existing reservation***" << endl;
cout << "Please enter the seat number which you want to cancel: ";
cin >> seatnum;
while (seatnum < 1 || seatnum > 15)
{
    cout << "Invalid input, please enter your selection again: ";
    cin >> seatnum;
}
if(seats_status[seatnum-1] == 1)
{
    cout << "The seat number " << seatnum << " is available now..." << endl;
    seats_status[seatnum - 1] = 0;
}
else
{
    cout << "The seat is not reserved" << endl;
}
```


Lab 9.3
NCU Theater Reservation System(I)

◆Step. 2（指令3 4）

```
void show_available() //instruction3
{
    cout << "***Show the all available seats***" << endl;

    for (int i = 0; i < 15; i++)
    {
        if (seats_status[i] == 0)
        {
            cout << "seat " << i+1 << endl;
        }
    }
    cout << endl;
    return;
}

void show_status() //instruction4
{
    cout << "***Show the status of all seats***" << endl;
    cout << "Seat number       Status" << endl;

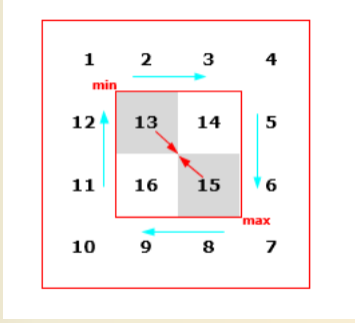
    for (int i = 0; i < 15; i++)
    {
        cout << "Seat " << setw(13) << left << i + 1 ;
        if (seats_status[i] == 0)
        {
            cout << "available" << endl;
        }
        else if (seats_status[i] == 1)
        {
            cout << "reserved" << endl;
        }
    }

    cout << endl;

    return;
}
```

Lab 9.4
Spiral Graph

- In this problem, the program will generate a spiral graph with numbers.



Lab 9.4
Spiral Graph

- First, the program will take a number to determine the dimension of the graph, then output the numbers from **top left to center** by clockwise, and the numbers follow increasing order.
- The range of the input dimension is [3, 31].
- The representation of each number needs to follow the center number. Ex. If the center number is the hundreds digit, each number must be presented in 3 digits, and the insufficient digits must be filled with 0.
- You need to use multi-dimension array or vector to store the numbers of the spiral graph, and the program will continue to execute until the user enters -1.

Lab 9.4
Spiral Graph

◆The screen dialog should appear as follows:

```
Input dimension: 3

1 2 3
8 9 4
7 6 5

Input dimension: 5

01 02 03 04 05
16 17 18 19 06
15 24 25 20 07
14 23 22 21 08
13 12 11 10 09

Input dimension: 10

001 002 003 004 005 006 007 008 009 010
036 037 038 039 040 041 042 043 044 011
035 064 065 066 067 068 069 070 045 012
034 063 084 085 086 087 088 071 046 013
033 062 083 096 097 098 089 072 047 014
032 061 082 095 100 099 090 073 048 015
031 060 081 094 093 092 091 074 049 016
030 059 080 079 078 077 076 075 050 017
029 058 057 056 055 054 053 052 051 018
028 027 026 025 024 023 022 021 020 019

Input dimension: -1

Process returned 0 (0x0)   execution time : 68.661 s
Press any key to continue.
```


Lab 9.4
Spiral Graph

◆思考方向：



Lab 9.4
Spiral Graph

◆Step. 1, 2

```
int matrix[50][50];

while(1)
{

    int dimension, i=0, j=0, x=1;
    cout << "Input dimension: ";
    cin >> dimension;
    if(dimension == -1)
        return 0;
    cout << endl;

    while(x <= dimension*dimension)
    {
        for (i = j; i < dimension - j; i++)
            matrix[j][i] = x++;
        for (i = j + 1; i < dimension - j; i++)
            matrix[i][dimension - j - 1] = x++;
        for (i = dimension - j - 2; i >= j; i--)
            matrix[dimension - j - 1][i] = x++;
        for (i = dimension -j - 2; i > j; i--)
            matrix[i][j] = x++;

        j++;
    }
```


Lab 9.4
Spiral Graph

◆Step. 3

```
if(dimension < 4)
{
    for (i = 0; i < dimension; i++)
    {
        for (j = 0; j < dimension; j++)
            cout << setw(1) << setfill('0') << matrix[i][j] <<" ";
        cout<<endl;
    }
}
else if(dimension < 10)
{
    for (i = 0; i < dimension; i++)
    {
        for (j = 0; j < dimension; j++)
            cout << setw(2) << setfill('0') << matrix[i][j] <<" ";
        cout<<endl;
    }
}
else if(dimension < 33)
{
    for (i = 0; i < dimension; i++)
    {
        for (j = 0; j < dimension; j++)
            cout << setw(3) << setfill('0') << matrix[i][j] <<" ";
        cout << endl;
    }
}
cout << endl;
```



Evaluation



- Lab9.1(30%)
  - 5 testcase, 6% each.
- Lab9.2(30%)
  - 5 testcase, 6% each.
- Lab9.3(30%)
  - 5 testcase, 6% each.
- Lab9.4(20%)
  - 5 testcase, 4% each.

